

Project Presentation of CO-OP Project at Industry (Module-2) (22CS421)

On

Artificial Intelligence: The Gap Between Expectations and Real-World Performance

Submitted By

Gurseerat Singh Sidhu
2210991599

Ritvik Sachdeva
2210990733

Riya Kundra
2210990735

Supervised By

Dr. Swati Goel
Assistant Professor (Zeta Cluster)

Department of Computer Science and Engineering,
Chitkara University, Punjab

- AI investment exceeded **\$180 billion** globally between 2024–2025.
- AI showed strong performance in reasoning, coding, and multimodal benchmarks.
- Organisations expected automation, faster workflows, and higher productivity.
- By 2026, many companies struggled with AI integration and measurable ROI.
- This mismatch between expectations and real-world performance is called the **AI Reality Gap**.
- Human supervision remains necessary due to AI errors and hallucinations.
- Poor-quality and unstructured enterprise data reduces AI reliability.
- Organisational resistance and workflow adaptation slow successful AI adoption.

The study uses two case studies to examine the AI Reality Gap.

The first case study, augmented Intelligence in Healthcare (Mayo Clinic) is mainly about:

- Diagnostic accuracy
- Clinical decision support
- Operational efficiency

The second case study, cross-model hallucination experiment using:

- OpenAI Chatgpt
- Google AI Mode.

Technical experiments conducted using Python and Google Colab.

UCI Heart Disease dataset used to study data quality degradation.

Random Forest model tested under missing, corrupt, and mixed noise conditions.

Productivity J-Curve mathematically modelled using Stanford AI Index and McKinsey data.

Research compares controlled benchmark performance with real-world deployment outcomes.

- Python programming language used for experiments and implementation.
- Google Colab used as the development and execution environment.
- Random Forest Machine Learning model used for data analysis.
- UCI Heart Disease Dataset used for clinical data experiments.
- OpenAI ChatGPT used for reasoning and hallucination testing.
- Google AI Mode used for cross-model evaluation.
- Mathematical modelling used for Productivity J-Curve analysis.
- Research references included Stanford AI Index 2025 and McKinsey AI reports.

- Two technical experiments were designed to study the AI Reality Gap.
- Experiment 1 analyzed how poor-quality data affects AI accuracy.
- Noise injection techniques used:
 - Missing value noise
 - Corruption noise
 - Mixed noise
- Random Forest model trained and tested at different noise levels.
- Experiment 2 implemented the Productivity J-Curve using mathematical modelling.
- Three adoption scenarios simulated:
 - Ideal adoption
 - Typical enterprise adoption
 - Proof-of-concept trap
- All experiments implemented using Python on Google Colab.
- Results compared with real-world case studies for practical analysis.

- Random Forest model achieved **85.25% accuracy** on clean clinical data.
- Accuracy dropped to **59.02%** under full mixed-noise corruption.
- Poor-quality enterprise data significantly reduced AI reliability and safety.
- Mayo Clinic achieved successful AI deployment through clean data and human oversight.
- Cross-model hallucination experiment showed AI can produce confident but incorrect responses.
- Productivity J-Curve showed organisational adoption strongly affects AI outcomes.
- Ideal adoption scenario reached productivity score of **132.0**, while failed adoption reached only **103.9**
- Research concluded that the AI Reality Gap is mainly a **systems and implementation problem**, not just a model problem.

- The AI Reality Gap is mainly caused by deployment and organisational issues, not lack of AI capability.
- Clean data and structured workflows are essential for reliable AI performance.
- Human oversight remains necessary in high-risk AI applications.
- Poor-quality enterprise data can significantly reduce AI accuracy and safety.
- Successful AI adoption depends on proper integration, governance, and long-term organisational investment.
- Explore Retrieval-Augmented Generation (RAG) systems to reduce hallucinations.
- Conduct long-term studies on AI adoption across enterprises
- Apply noise injection experiments to finance, legal, and manufacturing datasets.
- Study multi-model AI systems and chained reasoning failures
- Develop stronger human-in-the-loop verification frameworks for high-risk AI applications.

References



1. Stanford University Human-Centered AI, *Artificial Intelligence Index Report 2025*, Stanford University, Stanford, CA, USA, 2025. [Online]. Available: <https://aiindex.stanford.edu>
2. McKinsey & Company, *The State of AI in 2024*, McKinsey Global Institute, New York, NY, USA, 2024. [Online]. Available: <https://www.mckinsey.com/capabilities/quantumblack/our-insights/the-state-of-ai>
3. BARC Research, *Data and Analytics Trend Monitor 2025*, Business Application Research Center, Würzburg, Germany, 2025.
4. P. Nair *et al.*, "Artificial Intelligence–Electrocardiography to Predict Incident Atrial Fibrillation," *Mayo Clinic Proceedings*, vol. 97, no. 9, pp. 1638–1648, Sep. 2022.
5. OpenAI, *GPT-4 Technical Report*, OpenAI, San Francisco, CA, USA, 2023. [Online]. Available: <https://openai.com/research/gpt-4>
6. Google, *Gemini: A Family of Highly Capable Multimodal Models*, Google DeepMind, Mountain View, CA, USA, 2023. [Online]. Available: <https://deepmind.google/technologies/gemini>
7. M. Lichman, *UCI Machine Learning Repository: Heart Disease Dataset*, University of California, Irvine, School of Information and Computer Sciences, Irvine, CA, USA, 2013. [Online]. Available: <https://archive.ics.uci.edu/ml/datasets/heart+disease>
8. L. Breiman, "Random Forests," *Machine Learning*, vol. 45, no. 1, pp. 5–32, Oct. 2001.
9. E. Brynjolfsson, D. Rock, and C. Syverson, "Artificial Intelligence and the Modern Productivity Paradox: A Clash of Expectations and Statistics," in *The Economics of Artificial Intelligence: An Agenda*, A. Agrawal, J. Gans, and A. Goldfarb, Eds. Chicago, IL: University of Chicago Press, 2019, pp. 23–57.
10. D. Hendrycks *et al.*, "Humanity's Last Exam," *arXiv preprint*, arXiv:2501.14249, 2025. [Online]. Available: <https://arxiv.org/abs/2501.14249>

Thank You